Federal Spending Anomaly Detection

Project Summary

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This document summarizes the project: what it is, what was built, the key results, the main design decisions, and its limitations and next steps.

Repository: github.com/HarryAcharya/fed-spending-anomaly

Live dashboard: Tableau Public

1. The Project at a Glance

The project takes real, public federal contract award data and turns it into a short, ranked list of awards worth reviewing. It pulls NASA prime contract awards that were active in fiscal years 2022 and 2023 from the USASpending API, cleans and models the data in a PostgreSQL database, runs six documented rules that each look for one unusual pattern, gives every award a weighted risk score so the most concerning ones rise to the top, and presents the results in two dashboards, one in Power BI and one in Tableau.

The data and the analysis are real. The stakeholders and review team are simulated for a portfolio project, and that is stated openly throughout the work so no number is ever read as more than it is.

2. What We Built

Data pipeline

- A Python ingestion script that pulls awards from the USASpending API month by month, retries on failure, and removes duplicates using each award's permanent identifier.
- A raw staging table that holds the data exactly as it arrived, untouched, so the original is always recoverable.
- A cleaning step that standardizes names, parses the industry and product codes, and parses the dates into a separate clean table.

Database and data model

- A star schema built in PostgreSQL: one fact table of awards, plus dimension tables for recipient, industry code (NAICS), product or service code (PSC), and date.
- The date dimension is built on the federal fiscal year, which starts on the first of October, so reporting lines up with how the government actually counts time.
- Foreign keys link the fact table to every dimension, and the small number of missing links is measured and documented rather than hidden.

Anomaly detection

Six rules, each looking for a single pattern, each carrying a weight. An award's risk score is the sum of the weights of the rules it triggers, from zero up to a maximum of twelve. The score is then grouped into bands, None, Low, Medium, and High, and the exact rules that fired are written out in plain text on every award.

Rule	What it looks for	Weight	Awards flagged
R1	Award value above the 99th percentile of positive amounts, about 207 million dollars	3	167
R2	Award more than five times the recipient's own average, for recipients with three or more awards	3	291
R3	Amount sitting just under a federal competition threshold, near 250 thousand or 10 thousand	2	216
R4	Positive amount that is an exact multiple of one million dollars	1	13
R5	Recipient whose total awarded value is in the top one percent of all recipients	2	2,176
R6	End date before the start date, or a period of performance longer than ten years	1	255

Dashboards

- A Power BI report with two pages: an executive summary with the headline numbers and charts, and an anomaly review page that lists every flagged award with its reason and score.
- A Tableau Public dashboard built from the same data and published live, so anyone can open and explore it without any software.
- The two tools were built independently and show the same headline numbers, which is the cross check that the results are sound.

Business analysis documentation

A full set of documents that trace the project from a business need to the test that proves it: a project charter, a business requirements document, a functional requirements document, user stories, a data dictionary, the rule specifications, an accessibility review against Section 508 and WCAG 2.1 AA, a set of test cases, a requirements traceability matrix that links every requirement to a deliverable and a test, and a one page executive summary.

Quality and verification

- A seeded test that inserts a known award, runs the rules, and confirms only the expected rule fires, then rolls the test data back.
- A verification script that re-derives every documented figure straight from the live database and marks each one pass or fail. All twenty six checks pass, and the statistics, the risk band split, the full score distribution, and the flagship award all match the documents exactly.
- Everything is committed to a public GitHub repository with a README that explains the architecture, the results, how to reproduce it, and the honest limitations.

3. Key Results

The headline figures from the analysis.

Metric	Value
Total awards analyzed	16,866
Unique recipients	4,064
Industry codes (NAICS)	412
Product and service codes (PSC)	779
Total award value	222.93 billion dollars
Awards flagged by at least one rule	2,687 (about 16 percent)
High risk awards	64
Value held by the 64 high risk awards	about 108 billion dollars
Clean awards (no flag)	14,179
Risk score range	0 to 12 (highest observed was 9)
Highest risk award	Boeing, 22.4 billion dollar ceiling, score 9
Verification checks passed	26 of 26

In short, a small set of rules reduced 16,866 awards to a ranked list in which 64 high risk awards, about one fifth of one percent of the total count, hold roughly 108 billion dollars of value, the natural place for a reviewer to begin.

4. Key Design Decisions

These were the main design decisions and the reasoning behind each.

A star schema instead of one flat table. It keeps the data clean, makes joins simple, and is the standard shape that reporting tools like Power BI and Tableau expect, so the dashboards stay fast and easy to build.

Rule based detection with a weighted score, rather than a machine learning model. Every flag has a plain reason a reviewer can read and a business can defend. A black box that says an award is unusual without saying why is far harder to act on or to trust. The rules are transparent and auditable by design.

The large outlier rule used one global threshold, not a separate threshold per product code. The product code groups were often too small to give a stable threshold, which produced noise. A single, clearly stated percentile across all awards was more honest and more defensible.

The data is awards active during the window, not awards that started in it. The API matches awards that had activity in the window, so that is what the dataset honestly represents, and the documents say so plainly.

No agency dimension. Every award is NASA, so an agency column would only ever hold one value. Leaving it out was a deliberate modeling choice, not an oversight.

The award amount is the obligated and ceiling value, not money spent. This matters because the largest figure, the Boeing ceiling, is a contract maximum rather than cash out the door, and the project states this plainly.

Two dashboard tools instead of one. Building the same result in Power BI and Tableau both shows range across the two common tools and provides a cross check, since the two independently match to the dollar.

5. Limitations and Next Steps

Limitations

- The dataset is a single agency, NASA, so there is no agency dimension and no cross agency comparison.
- The award amount is the obligated and ceiling value, not money actually spent, so the largest figures are contract ceilings.
- The dataset is awards active during the window, not awards that started in it.
- The rules are deliberate heuristics chosen for clarity, not a statistical model, so they flag patterns for review rather than prove wrongdoing.
- The stakeholders and review team are simulated. The data and the analysis are real.

Next steps

- A short recorded video walkthrough of the project and dashboards.
- A statistical or machine learning layer on top of the rule based baseline, while keeping the readable rules as the foundation.
- Expansion beyond a single agency, which would bring back an agency dimension and allow comparison.
- An automated, scheduled refresh of the pipeline instead of a manual run.
- Full implementation of the accessibility recommendations, including a colorblind safe palette and complete keyboard navigation.
- Automated tests around the Python ingestion and cleaning code, alongside the database verification that already exists.

6. Technology Used

- PostgreSQL for the database and the star schema.
- SQL for the model and every rule, using joins, window functions, percentiles, and aggregation.
- Python for the API ingestion, cleaning, and loading.
- Power BI for the report, including measures and a two page design.
- Tableau Public for the live published dashboard.
- Git and GitHub for version control and publication.